Mwendwa Kiko

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EDUCATION

09/2022 - ongoing
PhD, Civil Engineering (Transportation Systems), University of Toronto (Canada)
Thesis: Modelling household automobile transactions for an Integrated Transportation and Land Use model.
Supervisor: Prof. Eric J. Miller

09/2020 – 12/2021
Masters in Transport and Sustainable Development, mention très bien (summa cum laude), École Nationale des Ponts et Chaussées (France)
Thesis: The Incorporation of Teleworking within Conventional Travel Demand Models: A Review of Past Studies and Initial Results with the Transport Model MODUS
Supervisor: Prof. Nicolas Coulombel

05/2013 – 09/2018
Bachelor of Science with First Class Honours, Civil Engineering, University of Nairobi (Kenya).

RESEARCH EXPERIENCE

09/2022 - ongoing
Doctoral Student, University of Toronto (Canada)
- Literature review of existing automobile ownership modelling methods.

03/2022 – 11/2022
Volunteer, Trufi Association (Germany)
- Carrying out interviews, drafting, editing and proofreading research papers.
- Co-author of a research paper published in Transportation Research Interdisciplinary Perspectives.

12/2021 – 09/2022
Research Engineer, Laboratoire Ville Mobilite Transport (France)
- Travel demand modelling of an experimental autonomous vehicle service under the SAM project for the Saclay region of France.
- Economic evaluation of the autonomous vehicle service.

06/2021 – 12/2021
Research Intern, Laboratoire Ville Mobilite Transport (France)
- Translation of the static, macroscopic, four-step transport model MODUS 3.1, previously written in SAS, to Python. Application of linear regression, non-linear regression, discrete-choice models and others.
- Extended the existing methodology for studying the effects of teleworking on mobility in the Greater Parisian region.
- First author on a research paper submitted to Transportation Research Record.

PROFESSIONAL EXPERIENCE

05/2023 – 10/2023
Engineering for Change Fellow, working with Bridges to Prosperity/Autodesk Foundation (Rwanda/USA).
- Analysis of different types of river crossing structures suitable for a rural transportation context.
- Development of a matrix tool that infrastructure planners in developing countries can use.

09/2018 – 08/2020
Graduate Engineer, Span Consulting Engineers LLP (Kenya)
- Civil and structural engineering design, project management and contract administration.

HONOURS AND AWARDS

$15,000	1st Place 2024-2025 IMI Big Data & Artificial Intelligence Competition, March 2025
$3,164	Alan Tonks Scholarship in Transportation Engineering awarded by the Department of Civil and Mineral Engineering at the University of Toronto, June 2024.
$500	1st Place, University of Toronto Mobility Five Minute Thesis (5MT) Competition, June 2024
$6,000	Arcadis | IBI Group Scholarship awarded by the Transportation Association of Canada (TAC), June 2024.
$105,000	NSERC Canada Graduate Scholarships — Doctoral, awarded by the Natural Sciences and Engineering Research Council of Canada, June 2024.
$10,000	Edith M. McCuaig-Throop Award for Black and/or Indigenous Students awarded by the University of Toronto School of Graduate Studies, July 2023.
$5,000	Roschlau Graduate Fellowship in Sustainable Urban Mobility awarded by the Mobility Network at the University of Toronto, May 2023.
$2,990	James Armstrong Civil Engineering Graduate Scholarship awarded by the Department of Civil and Mineral Engineering at the University of Toronto, March 2023
€10,500	Masters Scholarship awarded by the Embassy of France in Kenya and Somalia, July 2020.
$160	Best final year male student in Civil and Construction Engineering, University of Nairobi, September 2018.

PUBLICATIONS

First Author Publications in Refereed Journal

- **Kiko, M.,** Miller, E. J. (2025). Factors Influencing Car Ownership in Toronto: Insights from a Hurdle-Ordered Model. *International Journal of Sustainable Transportation*. *https://doi.org/10.1080/15568318.2025.2586219*
- **Kiko, M.,** Miller, E. J. (2025). Mobility Tools in Pre- and Post-COVID Toronto: Ownership Patterns and Model Transferability. *Travel Behaviour and Society (Under review).*
- **Kiko, M.,** Zhang, J., Zhang, Y., Miller, E. J. (2025). From Coast to Coast: Understanding Electric Vehicle Adoption Across Canadian Regions. *Transportation Research Part D (Under review)*
- **Kiko, M.,** Miller, E. J. (2026) Vehicle Prices in Microsimulation Models: Spatial Variation and Pandemic Effects. *Journal of Transportation Engineering, Part A: Systems (Under review)*
- **Kiko, M**., Coulombel, N., Poulhès, A, Seregina, T., Tremblin, G. (2023). Evaluation of Direct and Indirect Effects of Teleworking on Mobility: The Case of Paris. *Transportation Research Record, 2678, 3, 865-878, https://doi.org/10.1177/03611981231182973*

Middle Author Publications in Refereed Journals

- Hasselwander, M., **Kiko, M.**, & Johnson, T. (2022). Digital civic engagement, open data, and the informal sector: A think piece. *Transportation Research Interdisciplinary Perspectives*, *16*, 100700. https://doi.org/10.1016/j.trip.2022.100700

  Contribution: Conducting interviews, data analysis, revision of drafts.

- Eskenazi, M., Kaptue Bopda, F., **Kiko, M.,** Kotelnikova-Weiler, N., Tuncer, D., (2024) Scenario Co-design for Systemic Evaluation of Connected and Automated Mobility Setups. *IEEE Internet of Things Magazine, 7, 6, 36-42*

  Contribution: Literature review, draft preparation, revision of drafts.

Conference Presentations

- **Kiko, M.,** Jiaxin Zhang, Yu Zhang, Miller E. J., (6 – 10 July 2026) *From Coast to Coast: Understanding and Predicting BEV Adoption across Canadian Regions*, World Conference on Transport Research, Toulouse, France.

- **Kiko, M.,** Jiaxin Zhang, Jieru Zou, Miller, E. J., (7 – 11 June 2026) *How Far is Too Far? Online Vehicle Search Distances in Canada,* The 6th World Symposium on Transport and Land Use Research, Beijing, China.
- **Kiko, M.,** Miller, E. J., (5 – 9 January 2025) *Factors Influencing Car Ownership in Toronto: Insights from an Ordered Model*, TRB 104th Annual Meeting, Washington DC, USA.
- **Kiko, M.,** Miller, E. J., Haghighi, M., (14 – 18 July 2024) *The Interaction of Short-run Travel Behaviour and Mobility Tool Ownership,* 17th International Conference on Travel Behavior Research, Vienna, Austria.
- **Kiko, M.,** Miller, E. J. (12 – 15 May 2024) *EVs and Gas Stations: Insights from a Travel Demand Model*. 59th Annual Conference of the Canadian Transportation Research Forum, Kelowna, British Columbia.
- **Kiko, M.,** Coulombel, N. (8 – 10 June 2022) *Intégration du télétravail dans un modèle de demande de transport et évaluation socio-économique de ses effets : premiers résultats*. 4èmes Rencontres Francophones de Transports et Mobilité, Belval, Luxembourg.
- **Kiko, M.,** Coulombel, N. (17 – 18 March 2022) *Extent of the Effectiveness of Telework at a Metropolitan Scale: Lessons from a Travel Demand Model*. NECTAR Cluster 4 workshop in Marne-la-Vallée, Marne-la-Vallée, France.
- **Kiko, M.,** (15th October 2021) *The Future Of Transportation: Improving Road Safety Through Intelligent Transport Systems*. University of Nairobi 1st Annual Road Safety Conference, Nairobi, Kenya.

COMMITTEES AND LEADERSHIP

- Member, Peer-Review Committee (Comité de Suivi Téchnique), Ademe IT4Green Project (Teleworking Segment), France, 01/2025 – 01/2026.

RESEARCH INTERESTS

- Decarbonization of Global Transport Systems.
- ICT and transport.
- Public transport in the Global South.

PARTICIPATION IN UNIVERSITY LIFE

09/2023 – 08/2024 **Member of Executive Team,** University of Toronto Institute of Transportation Engineers (UT-ITE).

- Communications and Administration: In charge of media, keeping track of club membership and liaising with the University Administration.
- Coordinating a team of 7 undergraduate volunteers charged with social media work, website maintenance and development and logistics for our social events.

09/2023 – 08/2025 **Administrator**, Intelligent Transport Systems (ITS) Lab – SF3103.

- Lab security and access control.
- Enhancing the experience for lab users by coordinating with cleaners, maintenance and other staff.

TEACHING EXPERIENCE

09/2022 – ongoing **Teaching Assistant,** University of Toronto (Canada)

Probability Theory for Civil and Mineral Engineers under the supervision of Prof. Amer Shalaby.

Introduction to Transportation Systems under the supervision of Prof. Matthew J. Roorda (2 times)

- Performed all regular duties of a T.A. for a class size of 100 including coordinating with other teaching assistants, running tutorial sessions, office hours, and marking (course codes: CME263H1, CIV331H1).

02/2022 – **Teaching Assistant,** École Nationale des Ponts et Chaussées (France)

05/2022 Agent-Based Modeling and Simulation (ABMS) of Mobility and Transportation under the supervision of Dr. Tatiana Seregina.

- Prepared teaching materials, assisted students with the class project for a class size of 20.

COMPETENCIES

Linguistic
English – Native speaker
French – Fluent (Level C2)
Kiswahili – Fluent
Spanish – Beginner

Technical
Python
Statistical analysis
GIS (notably QGIS)
Travel demand modelling

COMPLEMENTARY TRAINING COURSES

04/2020 – 05/2020 **'Think Road Safety – Road Safety Training for External PARTNERS (facilitated)'** by the World Bank Group (The Open Learning Campus)

07/2018 **Experiencing China Summer School,** Tsinghua University (China)
Project: Exploring Wind as a Primary Source of Energy for China.

REFERENCES

Eric J. Miller
Professor
University of Toronto
eric.miller@utoronto.ca
(PhD supervisor)

Nicolas Coulombel
Associate Professor
École Nationale des Ponts et Chaussées
nicolas.coulombel@enpc.fr
(Supervisor during masters and previous employment)